CAPSTONE PROJECT

WEEK 1 – DESIGN DOCUMENT

PS-118: Court Case Listing and Adjournment Scheduling Engine

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Section	Computer Science and Design (CSD)
Domain	Public Services – Judiciary
Selected Language	**Python 3.10+**
Checkpoint	Week 1 – Design

Prepared for the 10 SECONDS Capstone Project

Design deliverables: glossary, class diagram, state diagram, business-rule register, decisions and test strategy

Command-file-driven engine. No graphical user interface is required.

1. Purpose and Scope

This document defines the Week 1 design for PS-118, a command-line engine that simulates administrative work in a district court registry. It reads bench setup, court holidays, case filings and hearing events from files, then produces admission decisions, bench assignments, daily cause lists, case diaries and pendency reports.

In scope: bench registration and allocation, limitation and wrong-forum exclusions, court-fee calculations, defective filings and cure deadlines, priority-based cause lists, board capacity, hearing outcomes, adjournments, stages, transfers, next dates, commercial disposal targets and reports.

Implementation: Python 3.10+ with modular services and deterministic outputs. The exact input schemas should be agreed by the team and adapted to the official sample files when received.

2. Glossary

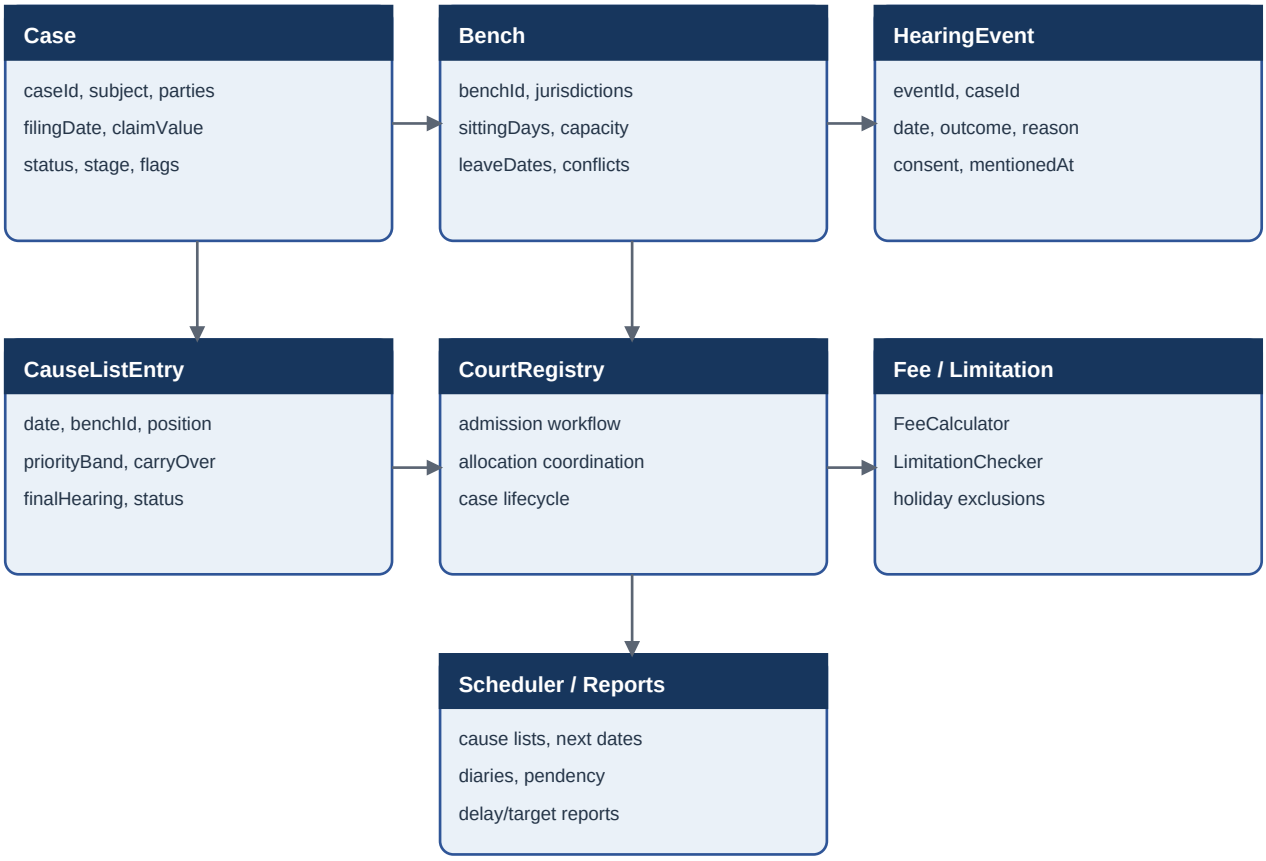
Term	Definition
Bench	A court unit with subject jurisdictions, sitting days, daily capacity, leave dates and conflict party IDs.
Case / filing	A matter submitted to the registry with subject, parties, cause-of-action/order date, claim value and urgency details.
Cause list	The ordered list of matters scheduled before a bench on a given sitting day.
Limitation period	The time allowed to file a case, appeal or revision under the project rules.
Wrong-forum exclusion	Qualifying time spent pursuing a matter in the wrong forum, excluded from limitation calculation.
Condonation	An application seeking acceptance of a filing made after its limitation deadline.
Court fee	The amount payable for a filing, calculated using the specified progressive bands and cap.
Defective filing	A filing with insufficient fee that must be corrected within the specified cure period.
Priority band	A category used to order matters for listing, from urgent/bail through other cases.
Carry-over	A matter not reached on a sitting day and carried to the next eligible sitting day.
Adjournment	A postponement of a hearing, with reason, count and possible cost.
Stage	A case lifecycle step: PLEADINGS, ISSUES, EVIDENCE, ARGUMENTS or JUDGMENT.
OVERBOARD	Flag used when the fallback scheduling rule places a matter on day 60 without available capacity.
Pendency	The set of unresolved cases, often summarized by age, stage and delay flags.
In camera	A private hearing; family cases involving a minor are listed only on designated days.
Command-file driven	The engine reads structured commands or data files and writes deterministic results, without a graphical UI.

3. System Components

Component	Responsibility
CommandProcessor / InputOutput	Read input files, validate required fields and write deterministic outputs.
Domain Models	Represent Case, Bench, HearingEvent, CauseListEntry, party IDs, stages and statuses.
AdmissionService	Apply limitation rules, wrong-forum exclusions, holiday extensions, fee checks and cure deadlines.
LimitationChecker	Calculate applicable deadline and decide whether condonation is required.
FeeCalculator	Apply progressive fee bands, cap and appeal fee reduction.
AllocationService	Filter benches by subject, commercial threshold, roster, leave and party conflicts; select by round-robin.
SchedulingService	Order priority bands, enforce daily capacity and final-hearing cap, and handle carry-over/urgent cases.
HearingService	Process outcomes, stage progression, adjournment counts/costs, transfers and next dates.
ReportingService	Produce cause lists, case diaries, pendency, evidence-delay and commercial-target reports.
CalendarService	Represent holidays, sitting days, bench leave and date calculations.

4. Class Diagram

The model separates case data from bench calendars, hearing events and generated cause-list entries. Services coordinate the business rules; input/output logic remains separate from domain logic.



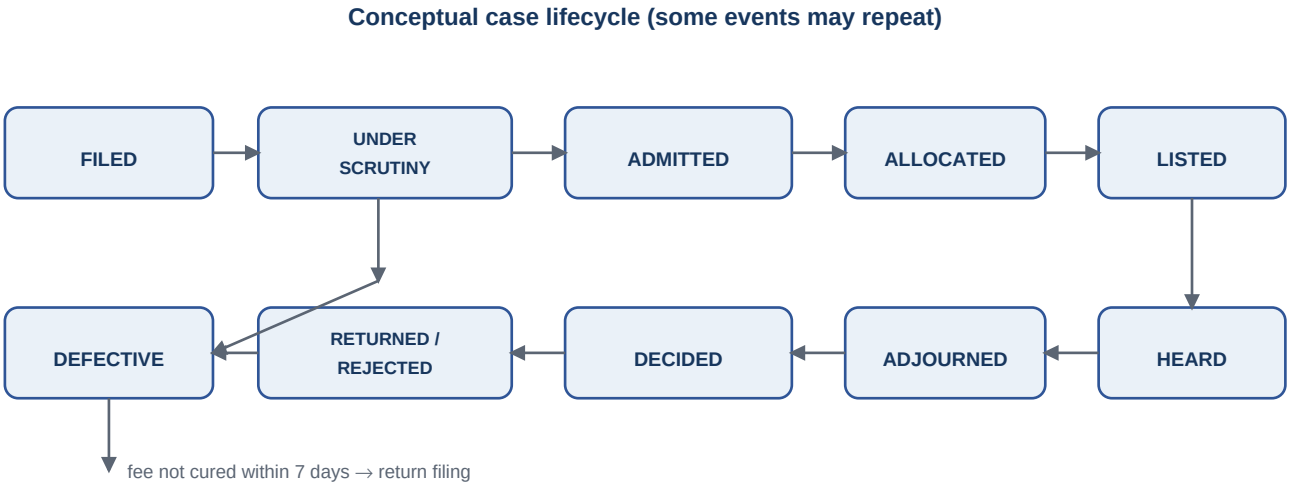
Key relationships: a case has one or more parties and hearing events; a bench handles many cases and has its own calendar; a cause-list entry links a case to a bench and date; services use the calendar, limitation and fee components. A transfer preserves the case's stage and history while changing its assigned bench.

Suggested core model fields

Class	Suggested fields
Case	case_id, case_type, subject, party_ids, filing_date, cause_of_action_date/order_date, claim_value, urgency, stage, status, adjournment_history, flags, assigned_bench_id, next_date
Bench	bench_id, subjects, sitting_days, daily_capacity, leave_dates, conflict_party_ids, designated_family_days, round_robin_position
HearingEvent	event_id, case_id, event_date, outcome, reason, requesting_party_id, consent_flag, mentioned_at, notes
CauseListEntry	case_id, bench_id, listing_date, priority_band, carry_over_count, final_hearing, position, overboard_flag
CalendarService	holidays, working-day checks, next eligible sitting day
AdmissionResult	case_id, decision, fee_due, fee_paid, cure_deadline, reason, flags

5. State Diagrams

5.1 Conceptual case lifecycle



Adjourned cases return to scheduling; transfer preserves stage/history.

The diagram is a high-level design. Actual transitions must follow the official rule register. A case may be listed and heard repeatedly; an adjournment returns it to scheduling. Transfer changes the assigned bench but preserves stage and history.

5.2 Stage progression

PLEADINGS → ISSUES → EVIDENCE → ARGUMENTS → JUDGMENT

A stage cannot be skipped unless the hearing event contains a consent flag. Reaching ARGUMENTS gives final-hearing priority, subject to the daily limit of eight final-hearing matters. A case in EVIDENCE for more than 180 days receives a DELAY flag and is included in the principal-judge report.

5.3 Admission and scheduling decision order

Order	Decision
Admission 1	Check limitation, including wrong-forum exclusion and holiday extension.
Admission 2	Check fee sufficiency; mark DEFECTIVE and set a seven-day cure deadline if deficient.
Admission 3	Check roster eligibility, then party conflicts and allocation availability.
Listing 1	Commercial disposal target overrides ordinary priorities; then urgent matters.
Listing 2	Apply carry-over priority bumps, then standard priority bands and oldest filing date first.
Listing 3	Apply daily capacity (40) and final-hearing cap (8), plus special calendar constraints.

6. Business-Rule Register (BR-01 to BR-51)

This register follows the supplied PS-118 brief. Module names are the planned design mapping; final implementation and test status should be updated as coding progresses.

Rule ID	Rule name	Requirement	Planned module
BR-01	Bench registration	Register each bench with subject jurisdictions, sitting days, daily capacity, leave dates and conflict party IDs.	Bench / Calendar
BR-02	Subject eligibility	Assign cases only to benches whose roster includes the case subject: civil, criminal, family or commercial.	Allocation
BR-03	Commercial threshold	Commercial claims over ■3 lakh must go to a commercial bench.	Allocation
BR-04	Party conflicts	Skip an otherwise eligible bench if its conflict list contains any party ID from the case.	Allocation
BR-05	Round-robin allocation	Distribute among eligible benches in round-robin order; maintain the next bench position per relevant roster/subject.	Allocation
BR-06	No eligible bench	If none is eligible, report the case as unallocated with a reason; document the chosen status.	Allocation / Decision log
BR-07	Limitation periods	Civil suits and money recovery: 3 years; appeals: 30 days from order; revisions: 90 days.	Admission
BR-08	Wrong-forum exclusion	Exclude supplied wrong-forum time when calculating limitation.	Limitation
BR-09	Holiday deadline	If the limitation deadline falls on a court holiday, extend it to the next working day.	Calendar / Limitation
BR-10	Late filing with condonation	A late filing is accepted only with a condonation application and is flagged DELAY_CONDONATION_PENDING.	Admission
BR-11	Late filing without application	Reject or return a late filing without a condonation application, with a clear reason; record selected status.	Admission / Decision log
BR-12	Admission order	Apply limitation (including exclusions and holiday extension) → fee sufficiency → roster → conflicts/allocation.	Admission
BR-13	Progressive fee bands	Apply 5% up to ■1 lakh; 4% on the portion above ■1 lakh up to ■10 lakh; 2% on the portion above ■10 lakh.	Fee calculator
BR-14	Fee cap	Cap the calculated fee at ■3 lakh.	Fee calculator
BR-15	Appeal fee	Appeals pay 50% of the original fee.	Fee calculator
BR-16	Deficient fee	If fee is deficient, register as DEFECTIVE and allow 7 days to cure.	Admission
BR-17	Uncured fee defect	If not cured within 7 days, return the filing.	Admission / Lifecycle
BR-18	Fee boundaries	Apply fee-band boundaries consistently at exactly ■1 lakh and ■10 lakh; document the progressive marginal-band interpretation.	Fee calculator / Decision log

6. Business-Rule Register (continued)

Rule ID	Rule name	Requirement	Planned module
BR-19	Priority bands	Highest first: (1) urgent/bail, (2) part-heard, (3) 3+ adjournments, (4) older than 2 years, (5) final hearing, (6) all others.	Scheduling
BR-20	Within-band order	Within a priority band, list oldest filing date first.	Scheduling
BR-21	Daily board capacity	Maximum 40 matters per bench per day.	Scheduling
BR-22	Final-hearing cap	Maximum 8 final-hearing matters per bench per day.	Scheduling
BR-23	Carry-over	Unreached matters carry to the next sitting day and go first within their band.	Scheduling
BR-24	Unreached escalation	A matter unreached on 3 consecutive sittings moves up one band.	Scheduling
BR-25	Commercial disposal target	Commercial cases past 12 months from filing are listed every 7 days until decided; this target overrides ordinary priorities.	Scheduling / Reporting
BR-26	Family/minor matters	Family cases involving a minor are heard in camera and listed only on designated days; designated days must be supplied or agreed as an assumption.	Scheduling / Decision log
BR-27	Urgent/bail mention	Mention before 10:30: same day if board has fewer than 40 matters; otherwise next sitting day at position 1.	Scheduling
BR-28	Priority precedence	Precedence: disposal target → urgent → carry-over bumps → standard bands → capacity caps.	Scheduling
BR-29	Hearing outcomes	Support HEARD, PART_HEARD, ADJOURNED, DECIDED and TRANSFERRED.	Hearings
BR-30	Case stages	Stages progress PLEADINGS → ISSUES → EVIDENCE → ARGUMENTS → JUDGMENT.	Lifecycle
BR-31	No stage skipping	Do not skip stages unless the event has a consent flag.	Lifecycle
BR-32	Arguments priority	Reaching ARGUMENTS gives final-hearing priority, still subject to the cap of 8.	Scheduling
BR-33	Evidence delay	More than 180 days in EVIDENCE triggers a DELAY flag and principal-judge report.	Lifecycle / Reporting
BR-34	Adjournment limit	Maximum 3 adjournments per party per stage; refuse the 4th unless reason is MEDICAL or COURT_OCCASIONED.	Hearings
BR-35	Court-occasioned adjournment	Court-occasioned adjournments do not count against a party.	Hearings
BR-36	Party adjournment cost	A party adjournment after 2 earlier adjournments costs ■2,000.	Hearings / Costs

6. Business-Rule Register (continued)

Rule ID	Rule name	Requirement	Planned module
BR-37	Commercial adjournment cost	Double the adjournment cost for commercial cases (■4,000).	Hearings / Costs
BR-38	Cost payment timing	Adjournment cost is payable before the next date.	Hearings / Costs
BR-39	Minimum interval to next date	Next date is the next eligible bench sitting day at least 14 days after adjournment, or 7 days for urgent cases.	Scheduling
BR-40	Calendar exclusions	Skip bench leave days and court holidays.	Calendar / Scheduling
BR-41	Capacity on next date	The selected next date must have board capacity.	Scheduling
BR-42	Overboard fallback	If no date has capacity within 60 days, list on day 60 and add OVERBOARD.	Scheduling
BR-43	Medical fourth adjournment	MEDICAL permits a fourth adjournment under the exception; whether it incurs a cost is unspecified and must be documented.	Hearings / Decision log
BR-44	Transfer limit and reason	Transfer a case at most once, and only because of roster change or conflict.	Transfer
BR-45	Destination eligibility	Check destination roster eligibility and conflicts against all case parties.	Transfer
BR-46	Preserve history on transfer	Preserve stage and adjournment history when transferring.	Transfer
BR-47	Transfer counts as adjournment	Count a transfer as an adjournment.	Transfer / Hearings
BR-48	Recompute after transfer	Recompute next date using the destination bench calendar, leave, holidays and capacity.	Transfer / Scheduling
BR-49	Cause-list report	Print daily cause lists by bench, showing order and carry-over where applicable.	Reporting
BR-50	Case diary	Include admission, fee, bench, stage, hearing history, adjournments/costs, flags and next date.	Reporting
BR-51	Pendency report	Include pending cases, age, stage, overdue commercial targets and evidence-delay flags.	Reporting

7. Assumptions and Decision Log

The source brief explicitly asks the team to document unresolved interpretations. Confirm these with the mentor where possible. If guidance is unavailable, record one consistent interpretation, its reason and an example.

ID	Question / topic	Action to record
DEC-01	3+ adjournments priority band	Decide whether the count is per case, per party or per stage.
DEC-02	Medical fourth adjournment cost	Clarify whether the permitted fourth MEDICAL adjournment incurs a cost; document the chosen interpretation.
DEC-03	Progressive court fee	Confirm marginal-band arithmetic and exact ■1 lakh / ■10 lakh boundaries.
DEC-04	Calendar format	Specify how court holidays, working days, sitting days and bench leave are represented.
DEC-05	Family/minor designated days	Obtain supplied designated days or document an agreed assumption.
DEC-06	Late filing without condonation	Choose the status/reason used for rejected or returned filing.
DEC-07	Transfer destination conflict	Define the result if the proposed destination bench conflicts with a party; do not silently bypass BR-45.
DEC-08	Urgent matter next day full	Clarify behavior if the next sitting day cannot fit the urgent matter despite the position-1 rule.
DEC-09	Exactly 14 days later	Confirm whether the 14th day is valid when it is an eligible sitting day with capacity.
DEC-10	Day-60 fallback	Clarify behavior if day 60 is a holiday or non-sitting day.
DEC-11	No eligible bench	Select and document the unallocated status and reason.

8. Edge Cases and Validation Checklist

Test case	Expected behavior from brief
Limitation deadline falls on court holiday	Extend deadline to next working day.
Claim value exactly ■1 lakh or ■10 lakh	Apply documented progressive fee-band boundaries consistently.
Fourth adjournment reason is MEDICAL	Permit the exception; apply documented cost interpretation.
Urgent mention at 10:29 when board has 40 matters	List next sitting day at position 1, subject to clarified capacity edge behavior.
Date exactly 14 days after adjournment is bench leave	Skip that date and find next eligible sitting.
Case unreached on exactly 3 consecutive sittings	Move it up one priority band.
Commercial case crosses 12 months while adjourned	Apply disposal-target priority and list every 7 days until decided.
Transfer destination bench conflicts with a party	Do not allocate to conflicted bench; apply documented transfer-failure behavior.

9. Initial Test Strategy

Use unit tests for individual rules and integration tests for complete command-file scenarios. Each test should identify the BR-ID, input, expected result and actual result. Include positive, negative and boundary tests.

Test group	Coverage
Admission / limitation	All limitation periods, wrong-forum exclusions, holiday extension, late filing and condonation.
Court fees	Progressive bands, ■1 lakh / ■10 lakh boundaries, cap, appeal reduction, defective filing and cure deadline.
Allocation	Subject mismatch, commercial threshold, conflicts, round-robin, no eligible bench.
Scheduling	Priority order, oldest-first ordering, board capacity 40, final-hearing cap 8, carry-over, three-sitting escalation, urgent mention.
Hearings / stages	All outcomes, stage progression, consent, adjournment limits, exceptions, costs and payment timing.
Next date / transfer	14-day and 7-day minimums, leave/holidays, capacity, day-60 OVERBOARD, transfer limit and destination conflict.
Reports	Cause lists, case diaries, pendency, evidence delay and commercial disposal targets.

10. Python 3.10+ Project Setup Recommendation

```
T070-PS-118/
README.md
requirements.txt
main.py
models.py
input_output.py
config.py
admission.py
allocation.py
scheduling.py
hearings.py
reporting.py
data/
benches.csv
holidays.csv
filings.csv
events.jsonl
tests/
test_admission.py
test_allocation.py
test_scheduling.py
test_hearings.py
test_reporting.py
docs/
week1-design.pdf
decision-log.md
business-rules.md
```

Use Python enums for fixed values such as case stages and hearing outcomes. Use date/datetime objects for date logic and Decimal for money calculations. Keep business rules in services and file parsing in the input/output layer. The precise input schema should be finalized against official sample files.

11. Week 1 Completion Checklist

- Glossary prepared.
- Class diagram and relationships prepared.
- Case lifecycle and stage progression diagrams prepared.
- Business-rule register BR-01 through BR-51 prepared.
- Edge cases and boundary conditions listed.
- Admission, priority and scheduling decision order documented.
- Assumptions and decision log prepared.

- Selected language recorded as Python 3.10+.
- Initial test strategy and repository structure documented.

Important: This is the Week 1 design artifact. It does not claim that any code or tests have already been implemented. Update the decision log after mentor guidance and revise schemas when the official sample data and expected outputs are shared.